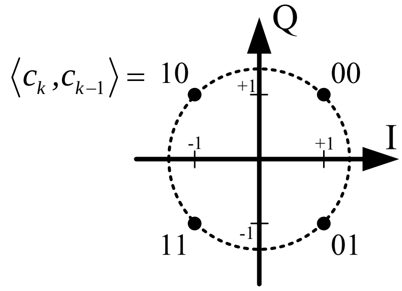
Project: Viterbi Decoder for MLSE

Consider a QPSK system, which discrete-time equivalent channel is given by $p_0 = 1,\ p_1 = 0.3$, and $p_2 = 0.7$, and which equivalent discrete-time noise is a discrete-time WSS Gaussian random process with

$$\mathrm{E}\left[w_k w_j^*\right] = \begin{cases} 1, & k = j \\ 0, & else \end{cases}$$

Assuming the constellation is as shown below.



1. First, sketch one section of the trellis diagram for this system. You should give a clear definition of your state vector and label a few "typical" branches.
2. Assuming that a sequence of QPSK symbols $x_{-2}, x_{-1}, x_0, \ldots, x_9, x_{10}, x_{11}$, is transmitted, where x_{-2}, x_{-1}, x_{10} and x_{11} are all known to be $(1 + j)$. The corresponding sequences of symbol-spaced received signal symbols are given by

$$\begin{bmatrix} y_0 \\ \vdots \\ y_{11} \end{bmatrix} = \begin{bmatrix} 0.0222 - j0.0169 \\ 1.4509 - j0.7471 \\ -1.6027 - j1.9480 \\ 1.6105 - j2.2362 \\ -1.1898 + j0.1263 \\ -0.6067 + j0.8870 \\ 0.0579 - j0.1223 \\ 0.6309 - j0.4483 \\ -0.0330 + j0.2217 \\ 1.5283 - j1.6817 \\ 0.4960 + j1.1453 \\ 2.3859 + j0.7010 \end{bmatrix}$$

3. Using MATLAB to find the transmitted symbols $x_0, \ldots, x_9$ based on the maximum likelihood sequence estimator. Please show your program and simulation result in your report and final presentation.